PHYS 580 Lab 12

Assignment: in this lab you will perform molecular dynamics simulations of a two-dimensional system of particles interacting via the so-called Lennard-Jones potential (cf. Chapter 9 in the textbook). You are provided the relevant starter program in Python (*md2.py*) or the codes in Matlab (*md2.m* for the main program, *updatemd.m* to perform one timestep, and *calc_energy.m* to calculate the current value of the energy).

1. Use the starter programs (or your own equivalent ones) to simulate a system of, say, 25 particles in a square box of side length 5 (in units of σ, the Lennard-Jones parameter). Initially, let the particles be at rest but with positions shifted by relatively small random amounts away from the evenly spaced, square lattice vertices. (Why is it good to include such position variations?) Then, as the simulation proceeds, produce images of the time evolution of particle positions similar to those displayed in Fig. 9.6 of the textbook. In addition, reproduce the time series of the total energy, temperature, tagged particle and tagged pair separations. Are the fluctuations in energy and temperature, and the trend of the pair separation as you expect, and why?

2. Find a way to speed up the convergence to equilibrium you observed in (1). In particular, use the feature of the starter program that allows one to change the kinetic energy of the particles via keyboard input during the simulation. Similarly, when you have attained a stable triangular arrangement of the particles (*solid*), find a way to *melt* it by heating the system. Demonstrate that you succeeded in melting the crystal by making appropriate plots of the particle arrangements and the time series of various functions.

   Note: discuss in your writeup what happens, and why, if the time step is too large (or too small) or if you raise the temperature by too much.

3. Study how varying the density, initial velocities and/or positions affects the approach to equilibrium and the nature of the final equilibrium configuration. You do not need to be exhaustive on this. For example, try putting 25 particles in a square of side length 10, and see how their characteristics change as you vary the temperature, substantiating your discussion with various time series graphs.